tf.compat.v1.math.log_sigmoid Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/compat/v1/math/log_sigmoid

> *Computes log sigmoid of x element-wise.*

Function Signatures

```
tf.compat.v1.math.log_sigmoid( x, name=None )
y = log(1 / (1 + exp(-x)))
y = -tf.nn.softplus(-x)
```

Code Examples

```
tf.compat.v1.math.log_sigmoid(
x, name=None
)
```

```
tf.compat.v1.math.log_sigmoid(
x, name=None
)
```

```
x = tf.constant([0.0, 1.0, 50.0, 100.0])
tf.math.log_sigmoid(x)
```

```
x = tf.constant([0.0, 1.0, 50.0, 100.0])
```

```
tf.math.log_sigmoid(x)
```

```
array([-6.9314718e-01, -3.1326169e-01, -1.9287499e-22,
-0.0000000e+00],
```

```
dtype=float32)>
```

Documentation

Computes log sigmoid of x element-wise.

Specifically, y = log(1 / (1 + exp(-x))). For numerical stability,
we use y = -tf.nn.softplus(-x).

Returns

Usage Example:

If a positive number is large, then its log_sigmoid will approach to 0 since
the formula will be y = log(/ (1 +)) which
approximates to log (1) which is 0.

If a negative number is large, its log_sigmoid will approach to the number
itself since the formula will be y = log(1 / (1 +)) which is
log (1) - log ((1 +)) which approximates to -
that is the number itself.